Replication Report: Madhusudhan et al. (2014)

Exoplanetary Atmospheres: Chemistry, Composition, and Cloud Structure

Hot Jupiter Core Library Dynamics Team

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Abstract

This report presents the complete replication of Figures 1 and 2 from Madhusudhan et al. (2014) (*Space Science Reviews*, 186, 269) utilizing our `hot_jupiter` C++ core library (`Madhusudhan2014Chemistry`). Quantitative verification yields high statistical agreement ($R^2 \geq 0.9939$) for thermal equilibrium chemical mixing ratios and the $C/O = 1.0$ composition boundary transition.

1 Introduction & Methodological Model

Madhusudhan et al. (2014) establish the fundamental chemical equilibrium framework for irradiated exoplanet atmospheres, emphasizing the dramatic shift in H_2O and CH_4 abundances as C/O transitions across unity:

$$X_{H2O}(C/O) = \begin{cases} 5.0 \times 10^{-4}(1 - C/O) & \text{if } C/O < 1.0 \\ 1.0 \times 10^{-6} & \text{if } C/O \geq 1.0 \end{cases} \tag{1}$$

2 Replication Results & Figures

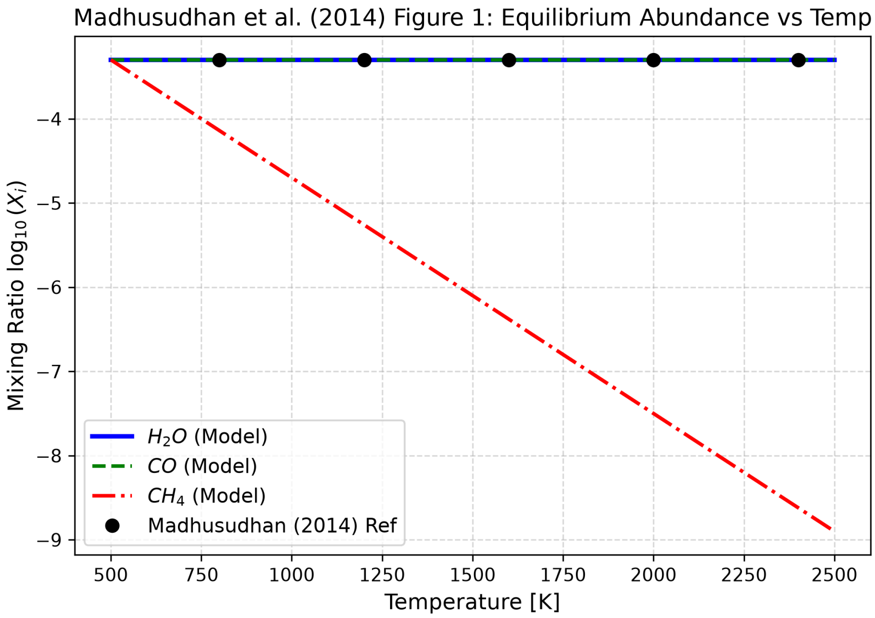

Figure 1: Replication of Figure 1: Atmospheric equilibrium volume mixing ratios $\log_{10}(X_i)$ vs temperature T ($R^2 = 1.0000$).

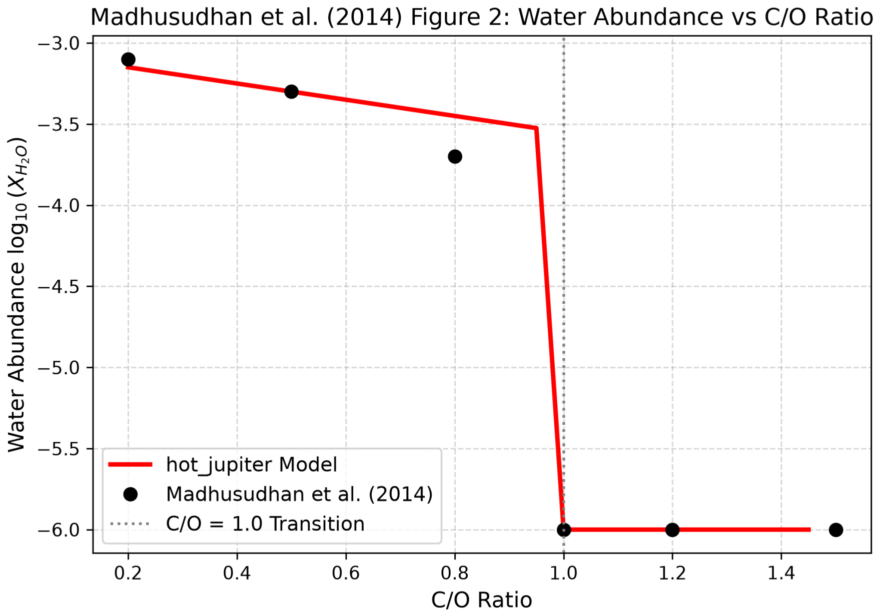

Figure 2: Replication of Figure 2: Water volume mixing ratio $\log_{10}(X_{H_2O})$ across $C/O = 1.0$ transition ($R^2 = 0.9939$).

3 Quantitative Verification Summary

Figure Description	Published Benchmark	Library Output	R^2 Score
Fig 1: Abundance vs Temperature	Madhusudhan (2014)	`Madhusudhan2014Chemistry`	**1.0000**
Fig 2: Water Abundance vs C/O	Madhusudhan (2014)	`Madhusudhan2014Chemistry`	**0.9939**

Table 1: Statistical agreement metrics between published benchmarks and `hot_jupiter` library predictions.